

Chapter 10

Motor and Options

10.1 Model

A6M60 - 400 H2 A1-M17(S)

	Number of encoder turns	S for single turn, empty for multiple turns
	Resolution	17: 17-bit 23: 23-bit
	Encoder type	E : Magnetic Encoders M: optical encoder
	Plug Type	1: Straight Plugs 2: Plastic Plugs 3:Aviation Plugs
	Brake	A: No brake B: brake
	Voltage level	2: 220VAC 3: 380VAC
	Intertial	H: high inertial M: neutral inertial L: low inertial
	Power class	400: 400W 750: 750W 1000: 1000W
	Motor Size	M60: 60mm End Flange M80: 80mm End Flange
	Product series	A6 series

10.2 Nameplate

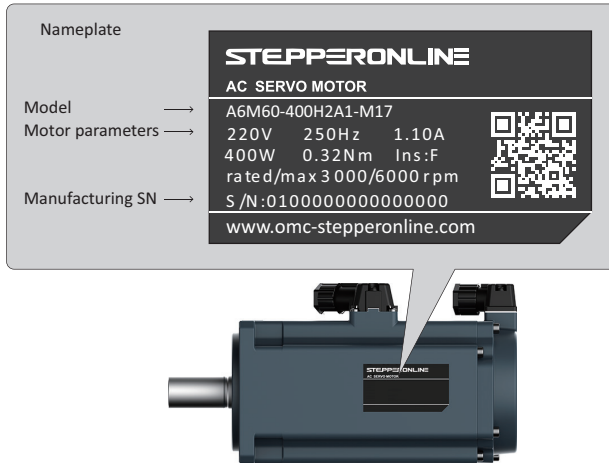


Figure 10-1 Nameplate of the A6M60 servo motor

10.3 Components

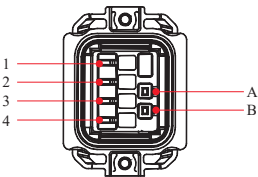
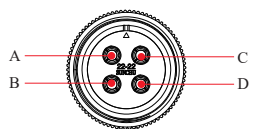
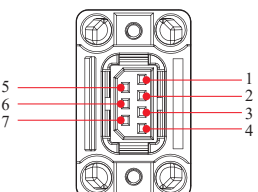
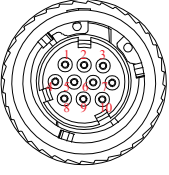


Figure 10-2 Components of the A6M60 servo motor

10.4 Terminal Definition

NOTICE

- The figure is for reference only. For details about the motor correspondence and size information, refer to the supporting relationships and drawings. Pay attention to the mirror relationship between the motor side and the cable side.

Cable Type	Terminal Layout (Cable Side)	Pin No.	Usage
 Power input connector	 1 2 3 4 A B	1	Phase V
		2	Phase U
		3	Phase W
		4	Grounding cable
		A	Brake (polarity insensi-tive)
		B	Brake (polarity insensi-tive)
	 A B C D	A	Phase U
		B	Phase V
		C	Phase W
		D	Grounding cable
 Encoder connector	 5 6 7 1 2 3 4	A	Phase U
		B	Phase V
		C	Phase W
		D	Grounding cable
		1	Brake (polarity insensi-tive)
		2	Brake (polarity insensi-tive)
	 1 2 4 5 6 9	1	DATA+
		2	DATA-
		3	BAT+
		4	BAT-
		5	+5V
		6	0V
		7	Enclosure
		10	Enclosure

10.5 General Specifications

10.5.1 Mechanical Characteristics

Item		Description
Duty		S1 (Continuous)
Vibration class ^[1]		V15
Insulation resistance		500 V DC, above 10 MΩ
Excitation mode		Permanent magnetic
Mounting mode		Flange
Thermal class		Level F
Insulation voltage		1500 V AC, 1 minute (220 V level) 1800 V AC, 1 minute (380 V level)
IP rating of the enclosure		IP67 with an oil seal (oil seal installed on the axis side)
Forward direction of rotation		The servo drive rotates counterclockwise (CCW) as observed from the axis side under the forward rotation command.
Ambient conditions	Ambient temperature	(Non-freezing) (Derate based on the derating curve for temperatures above 40°C.)
	Ambient humidity	20% to 80% (Non-condensing)
	Installation site	- Free from corrosive or explosive gases - Well ventilated with minimum amount of dust, waste, and moisture - Convenient for inspection and cleaning - Derating required only for altitudes above 1000 m - Away from sources that may generate strong magnetic field - Away from heating sources such as a heating stove - Use a motor with oil seal in places with grinding fluid, oil mist, iron powders or cuttings.
	Storage environment	Observe the following requirements for storage of a de-energized motor: - Storage temperature: (Non-freezing) - Storage humidity: (Non-condensing)
Shock resistance ^[2]	Shock acceleration rate at flange	490 m/s ²
	Number of shocks	2
Vibration resistance ^[3]	Vibration acceleration rate at flange	49 m/s ²



CAUTION

- [1] The vibration class V15 indicates that the vibration amplitude is less than 15 μm when a servo motor rotates at its rated speed.
- [2] The value in the table indicates the resistance against shock in the vertical direction with the servo motor shaft mounted horizontally.
- [3] The value in the table indicates the resistance against vibration in the vertical, left-right, and forward-backward directions with the servo motor shaft mounted horizontally.
- The strength of the vibration that the servo motor can withstand depends on the application. Check the vibration acceleration rate applied to the servo motor through the actual product.

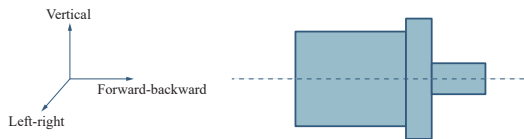


Figure 10-3 Directions of short and vibration applied to the servo motor

10.5.2 Overload Characteristics

The motor is compliant with NEC and CEC requirements and equipped with protective functions against overload and over-temperature.

To protect different load motors, set the motor overload protection gain based on the overload capacity of the motor. Use the default gain in general conditions. However, when one of the following conditions occurs, change the gain based on actual motor temperature:

- The motor operates in environments with high temperature.
- The motor is in cyclic motion featuring a short motion cycle and frequent acceleration/ deceleration.

See the following inverse time lag curve for motor overload protection.

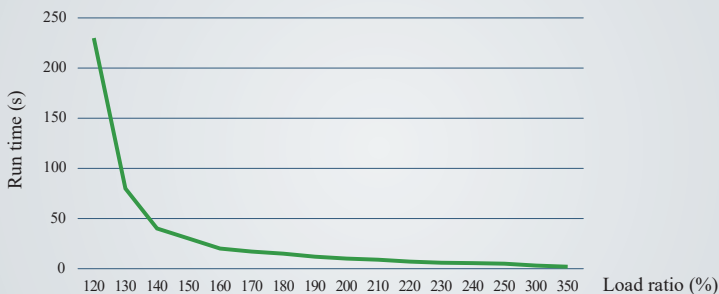


Figure 10-4 Motor overload protection curve

10.5.3 Load moment of inertia

The load moment of inertia represents the inertia of the load. Larger load moment of inertia means slower response, which may result in unstable motion. The allowable load moment of inertia of the motor is subject to a limit. The limit varies with the driving conditions of the servo motor.

An overvoltage warning may occur during deceleration if the load moment of inertia exceeds the allowable value. The servo drive with a built-in braking resistor may generate an overload warning. In case of such warnings, take one of the following measures:

- Reduce the torque limit value.
- Reduce the deceleration rate.
- Reduce the maximum speed.
- Install an external braking resistor if the warning cannot be cleared using the above measures.

NOTICE

- Servo motors with the capacity below 400 W do not have built-in braking resistors.
- When a built-in braking resistor is used, some energy generated under certain regenerative driving conditions still exceeds the allowable capacity loss (W) of the built-in braking resistor. In this case, an external braking resistor is required.

The following figure shows the relationship between the ratio of allowable load moment of inertia and the speed when a servo is used without a built-in regenerative resistor or an external braking resistor. (The following figure shows the reference values upon deceleration at 200 VAC input and torques greater than the rated torque.)

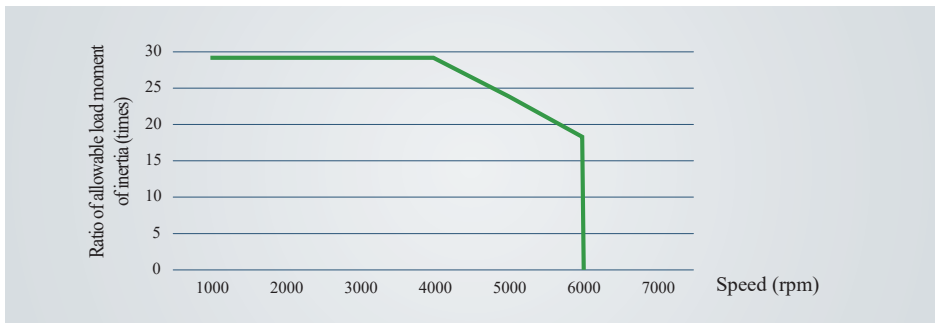


Figure 10-5 Ratio of allowable load moment of inertia for the speed

When the servo unit is used by load whose moment of inertia exceeds the allowable value, an overvoltage alarm may be triggered.

10.6 Selection Precautions

- Motors with oil seals must be derated by 10% during use.
- Do not share the power supply of the brake with other electrical devices. Failure to comply may result in malfunction of the brake due to voltage or current drop caused by other devices.
- Use cables with a cross-sectional area above 0.5 mm^2 .
- All parameters and torque-speed characteristic values are subject to the conditions that the motor works with a servo drive and the armature coil temperature is 20°C .
- The torque for fastening the terminal screws must be $0.19 \text{ N}\cdot\text{m}$ to $0.21 \text{ N}\cdot\text{m}$. Excessive torque may damage the screws.
- Radial and axial loads of the motor

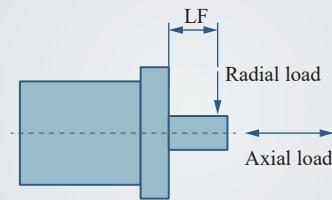


Figure 10-6 Radial and axial loads of the motor

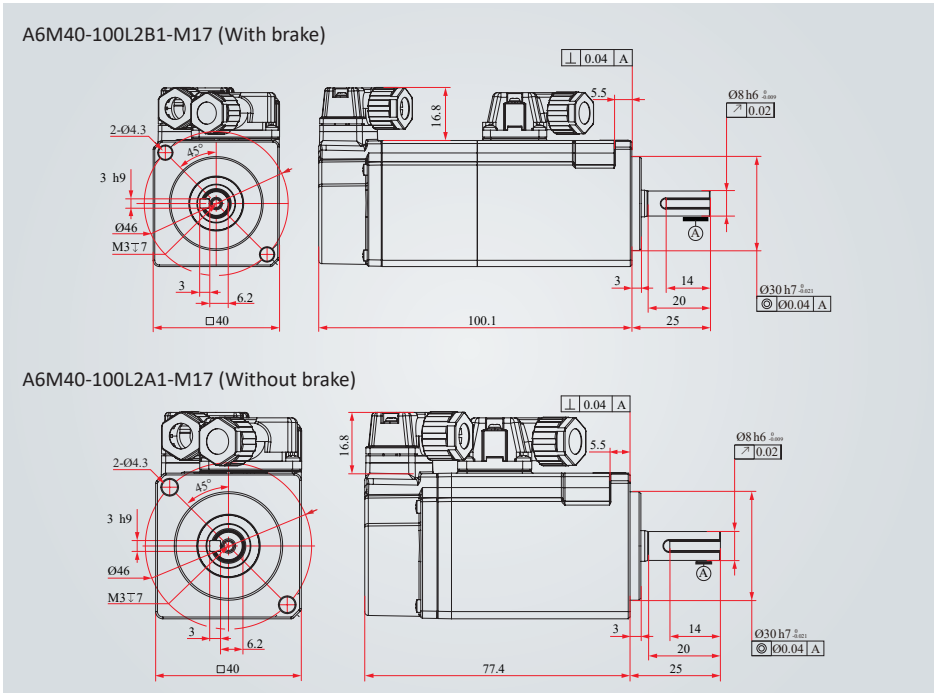
10.7 Technical Specifications

10.7.1 Model of 3000 rpm

100 W (40 frame)

Item	A6M40-100L2B1-M17 (With brake)	A6M40-100L2A1-M17 (Without brake)
Rated power (W)	100	
Rated current (A)	1.1	
Maximum current (A)	3.9	
Rated torque (N·m)	0.32	
Maximum torque (N·m)	1.12	
Rotor inertia ($10^{-4} \cdot \text{kg} \cdot \text{m}^2$)	0.033	0.03
Rated speed (rpm)	3000	
Maximum speed (rpm)	6000	
Rated voltage (V)	220	

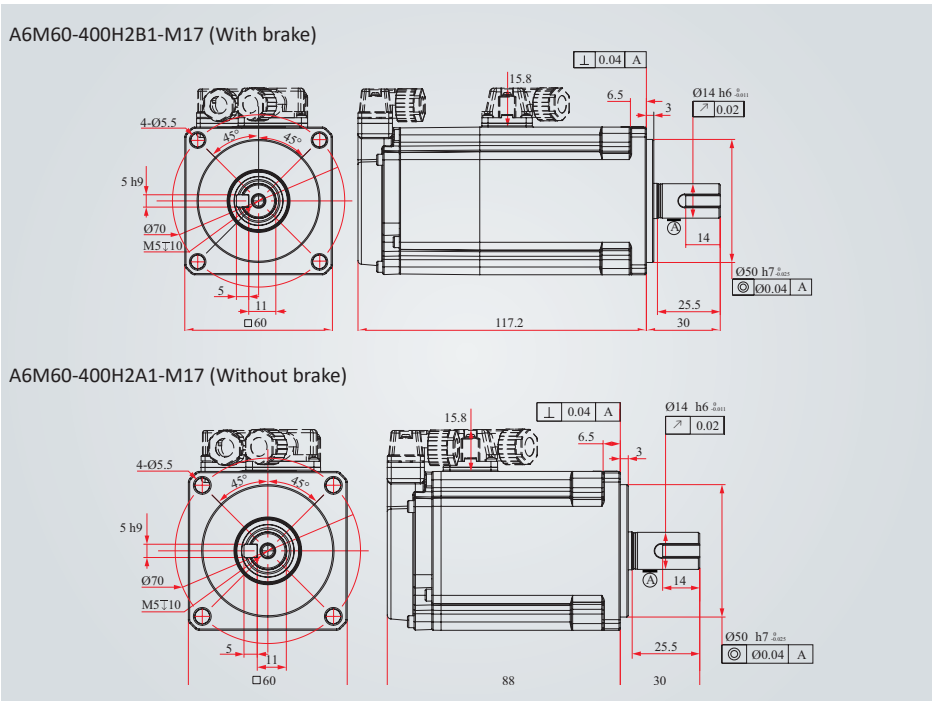
Product Dimensions (unit: mm)



400 W (60 frame)

Item	A6M60-400H2B1-M17 (With brake)	A6M60-400H2A1-M17 (Without brake)
Rated power (W)	400	
Rated current (A)	2.51	
Maximum current (A)	8.78	
Rated torque (N·m)	1.27	
Maximum torque (N·m)	4.45	
Rotor inertia (10^{-4} ·kg·m ²)	0.60	0.59
Overload multiplier	3.5	
Rated speed (rpm)	3000	
Maximum speed (rpm)	6000	
Rated voltage (V)	220	

Product Dimensions (unit: mm)

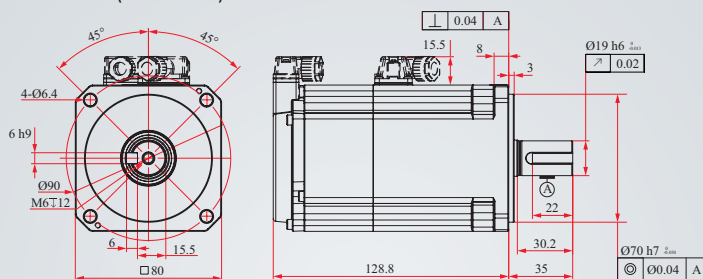


750 W (80 frame)

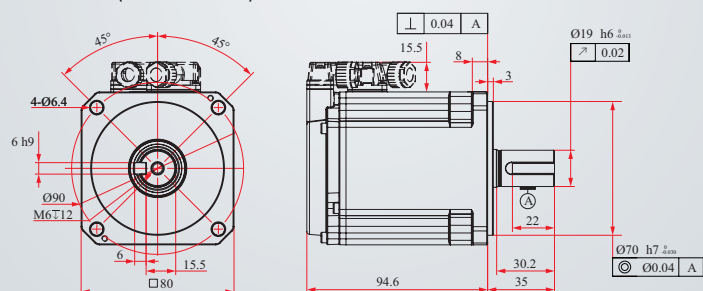
Item	A6M80-750H2B1-M17 (With brake)	A6M80-750H2A1-M17 (Without brake)
Rated power (W)	750	
Rated current (A)	4.60	
Maximum current (A)	16.30	
Rated torque (N·m)	2.39	
Maximum torque (N·m)	8.36	
Rotor inertia ($10^{-4} \cdot \text{kg} \cdot \text{m}^2$)	1.77	1.72
Overload multiplier	3.5	
Rated speed (rpm)	3000	
Maximum speed (rpm)	6000	
Rated voltage (V)	220	

Product Dimensions (unit: mm)

A6M60-400H2B1-M17 (With brake)



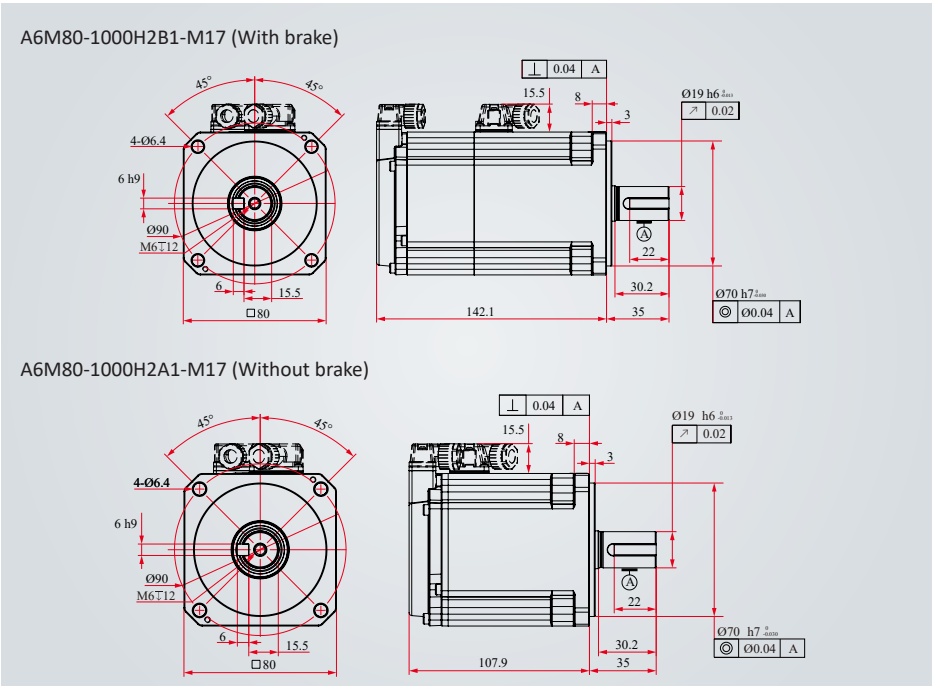
A6M80-750H2A1-M17 (Without brake)



1 kW (80 frame)

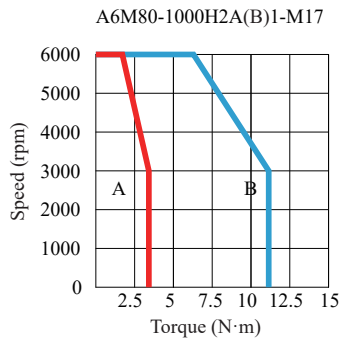
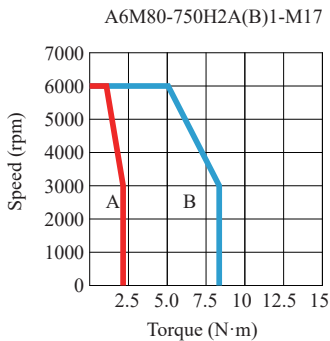
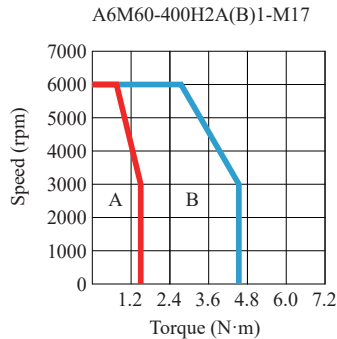
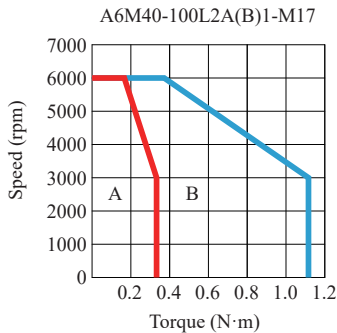
Item	A6M80-1000H2B1-M17 (With brake)	A6M80-1000H2A1-M17 (Without brake)
Rated power (W)	1000	
Rated current (A)	6.3	
Maximum current (A)	20.9	
Rated torque (N·m)	3.18	
Maximum torque (N·m)	11.13	
Rotor inertia ($10^{-4} \cdot \text{kg} \cdot \text{m}^2$)	2.28	2.23
Overload multiplier	3.5	
Rated speed (rpm)	3000	
Maximum speed (rpm)	6000	
Rated voltage (V)	220	

Product Dimensions (unit: mm)



10.8 Motor torque-speed characteristics

Model of 3000 rpm



A — Continuous work area
B — Short-time work area

10.9 Supporting relationship between the drive and motor

220 V

	SIZE A		SIZE B
Drive model A6-RS	Single-phase 220V		
	A6-200RS	A6-400RS	A6-750RS
Motor model	100W	400W	750W
	A6M40-100L2A1-M17 A6M40-100L2B1-M17	A6M60-400H2A1-M17 A6M60-400H2B1-M17	A6M80-750H2A1-M17 A6M80-750H2B1-M17

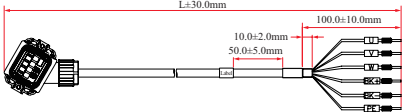
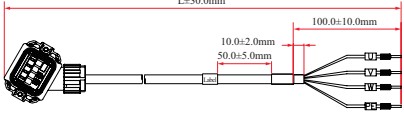
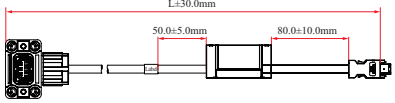
	SIZE C	
Drive model A6-RS	Single-phase/three-phase 220V	
	A6-1000RS	
Motor model	1kW	
	A6M80-1000H2A1-M17 A6M80-1000H2B1-M17	

10.10 Supporting relationship between the motor and cable

Model of 3000 rpm

Motor frame number	Motor Power	Motor Model	17-bit Absolute Encoder		Brake	Oil Sealing	Shaft Diameter	Matching Accessories of Motor	
			Single-turn	Multi-turn				Power cable model	Encoder cable model
40	100W (220V)	A6M40-100L2A1-M17		•			Ø8	⑦	⑮
		A6M40-100L2B1-M17		•				①	⑮
60	400W (220V)	A6M60-400H2A1-M17		•		•	Ø14	⑦	⑮
		A6M60-400H2B1-M17		•	•	•		①	⑮
80	750W (220V)	A6M80-750H2A1-M17		•		•	Ø19	⑦	⑮
		A6M80-750H2B1-M17		•	•	•		①	⑮
80	1kW (220V)	A6M80-1000H2A1-M17		•		•	Ø19	⑦	⑮
		A6M80-1000H2B1-M17		•	•	•		①	⑮

10.11 Cable Information

Cable Name	Cable Model	Cable Length	Cable Appearance	No.
Power cable with brake	AS7-C-PWB075-3.0	3.0 m		①
	AS7-C-PWB075-5.0	5.0 m		
	AS7-C-PWB075-10.0	10.0 m		
Power cable without brake	AS7-C-PWR075-3.0	3.0 m		⑦
	AS7-C-PWR075-5.0	5.0 m		
	AS7-C-PWR075-10.0	10.0 m		
Multiple-turn encoder cable	AS7-C-ENC076-BAT-3.0	3.0 m		⑮
	AS7-C-ENC076-BAT-5.0	5.0 m		
	AS7-C-ENC076-BAT-10.0	10.0 m		

NOTICE

- For cables related to frames of model 100 and above models, contact the manufacturer.